JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Affiliated to DBATU, RTMNU & MSBTE, Mumbai

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An Autonomous Institute, with NAAC "A" Grade

Education to Eternity

॥ज्ञानम् सर्वार्थ साधनम्॥

Department of Computer Science & Engineering

"A place to learn; A chance to grow"

Session: 2024-25

MID SEMESTER EXAMINATION 1

Course: B. Tech. in Computer Science & Engineering **Year/Semester:** 2nd year/ 4th Sem

Subject : Design & Analysis of Algorithm **Subject Code:** CS4T002

Max Marks: 20 **Date:** 27/02/2025 **Duration:-** 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.

2. Assume suitable data whenever necessary.

3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

CO1: Recall basic concepts of algorithm in analysis and Design of algorithms.

CO2: Examine Recurrence relations, solutions of recurrence of searching sorting methods.

CO3: Apply Greedy methods used for analysis and Design of Algorithm.

CO4: Analyse Dynamic Programming concepts in designing algorithm.

CO5: Evaluate advanced techniques and tools available for algorithm analysis and development.

SECTION A

3. All Questions are compulsory.

4. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Write down the best case, average case and worst case complexity of successful and unsuccessful search.	L1/CO2	1
Q.2	Recall the concept of Quick Sort.	L1/CO2	1
Q.3	State the principle of Job Sequencing with Deadline.	L1/CO3	1
Q.4	Define Spanning tree with example.	L1/CO3	1
	SECTION B **3. All Questions are compulsory.** **4. Answer in two or three sentence.**		
Q.5	State the concept of EPL and IPL. Discuss relation between ELP and IPL.	L1/CO2	2
Q.6	Describe the three main steps in a typical Divide & Conquer algorithm.	L2/CO2	2
Q.7	Explain the Minimum cost spanning tree with example.	L2/CO3	2
Q.8	List out the three methods of solving knapsack problem.	L1/CO3	2
	SECTION C **2. Solve any two questions.**		
Q.9	Find out average number of comparisons for successful and unsuccessful search for given array. 10, 20, 30, 40, 50, 60, 70, 80	L3/CO2	4
Q.10	Identify the best possible sequence for the job with deadline. Jobs: 1, 2, 3, 4, 5, 6, 7 Profit: 3, 5, 18, 20, 1, 6, 30 Deadline: 1, 3, 4, 3, 2, 1, 2	L2/CO3	4
Q.11	Design Huffman encoding for given file. FILE = { a:20, b:15, c:5, d:15, e:45 }	L5/CO3	4